

HAROKOPIO UNIVERSITY

SCHOOL OF DIGITAL TECHNOLOGY
DEPARTMENT OF INFORMATICS AND TELEMATICS

POSTGRADUATE PROGRAMME INFORMATICS AND TELEMATICS
COURSE WEB TECHNOLOGIES AND APPLICATIONS

Secure Location Sharing Platform for Closed Groups Using Home Assistant and Encrypted VPN Connection

Anastasios Kotronis

Examining Committee

Dimopoulos Alexandros (Supervisor)
Assistant Professor, Department of Informatics and Telematics, Harokopio University

Tsadimas Anargyros (Examiner)
Teaching Laboratory Staff, Department of Informatics and Telematics, Harokopio University

Dimitrakopoulos Georgios (Examiner)
Professor, Department of Informatics and Telematics, Harokopio University

Why This Project?

Core Idea: Location sharing

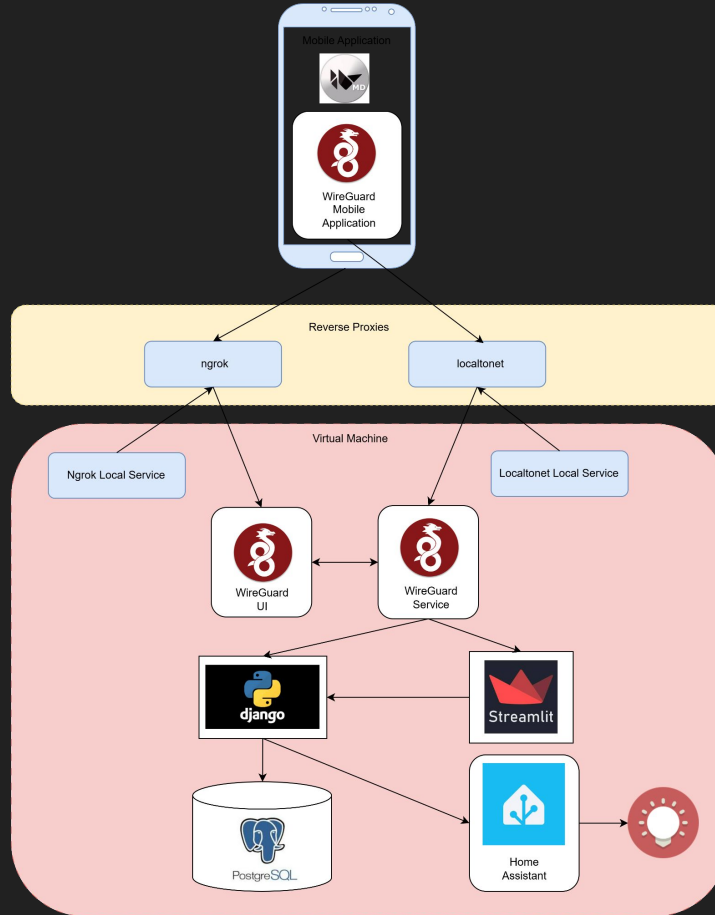
- Trend. Integrated in a wide range of applications
- Is often centralized and privacy-invasive
- Existing tools (e.g., Google Maps, HA) are not suitable for closed, self-managed groups
- Need for a private, self-hosted, VPN-protected alternative
- Real-time location triggers to automate smart devices
- User retains full control of data, network, and infrastructure

What will we present?

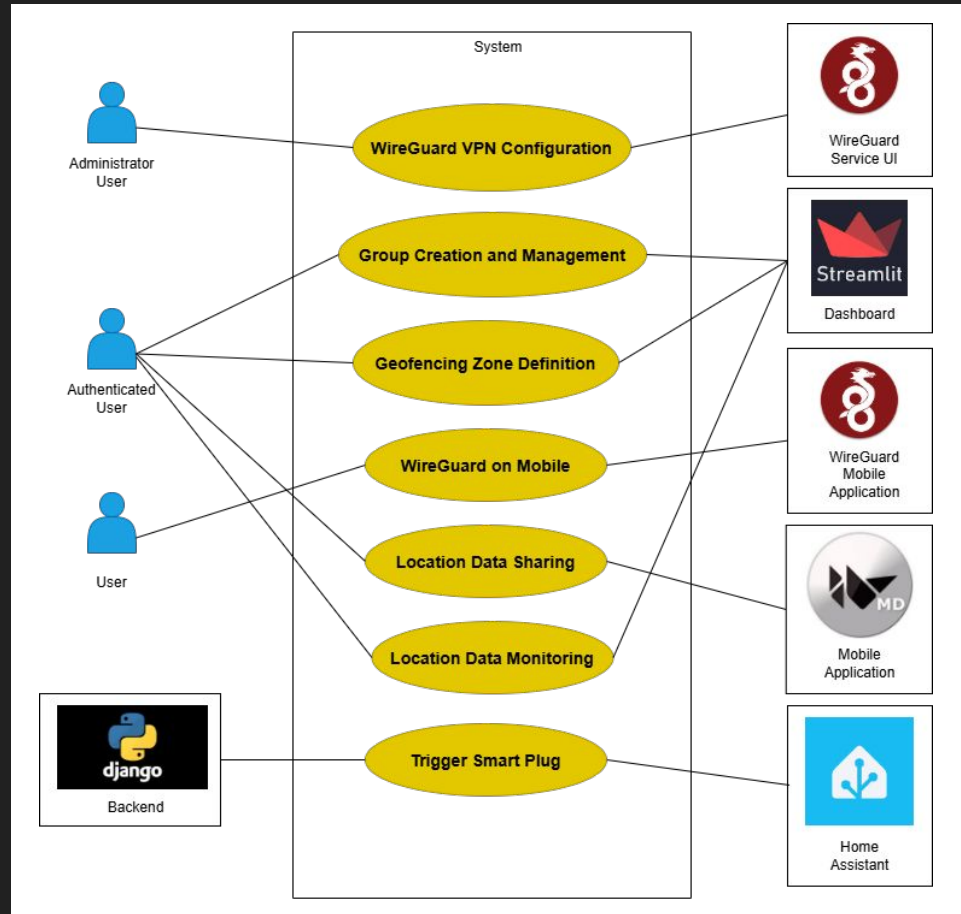
- A fully working prototype: mobile $\Rightarrow$ backend $\Rightarrow$ automation + front (monitoring)
- VPN-secured location sharing via WireGuard
- Group management with invitations
- Zone-based logic with smart device triggers and notifications
- Web-based monitoring dashboard
- Dockerized VM setup
- No port forwarding needed
- Live demo of main features

System Architecture

- Mobile browser ⇒ Ngrok ⇒ WG UI
- Mobile Kivy app ⇒ WG mobile ⇒ Localtonet ⇒ WG
- WG ⇒ routing ⇒ Django, Streamlit
- Django ⇒ PostgreSQL
- Streamlit ⇒ Django
- Django ⇒ Home Assistant
- Home Assistant ⇒ Tapo plug



Use Case diagram



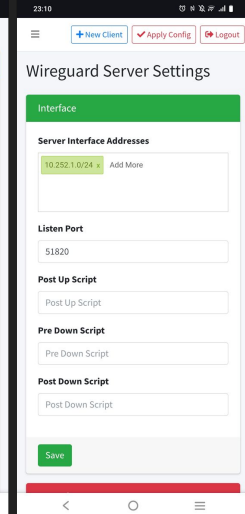
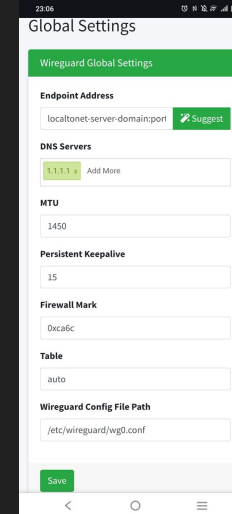
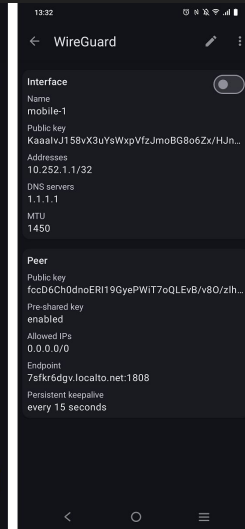
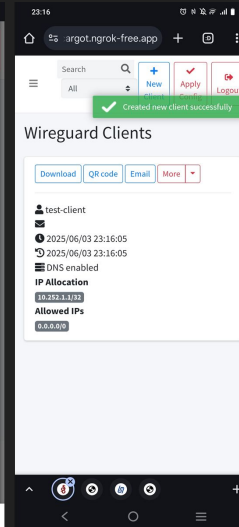
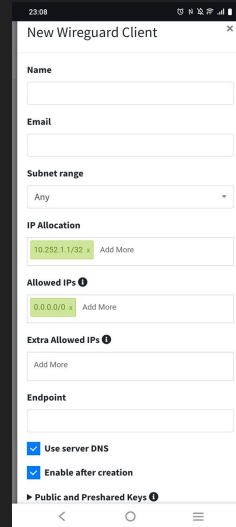
Reverse Proxies

Access self-hosted services

- Solution 1: Port forwarding
 - Network configuration dependence
 - Network provider dependence
- Solution 2: Reverse proxies
 - Avoid above dependencies
 - Secure communication
 - https for WireGuard UI through Ngrok
 - Encrypted for mobile to backend (WireGuard, Localtonet)
 - Third party dependence / flexibility trade-off

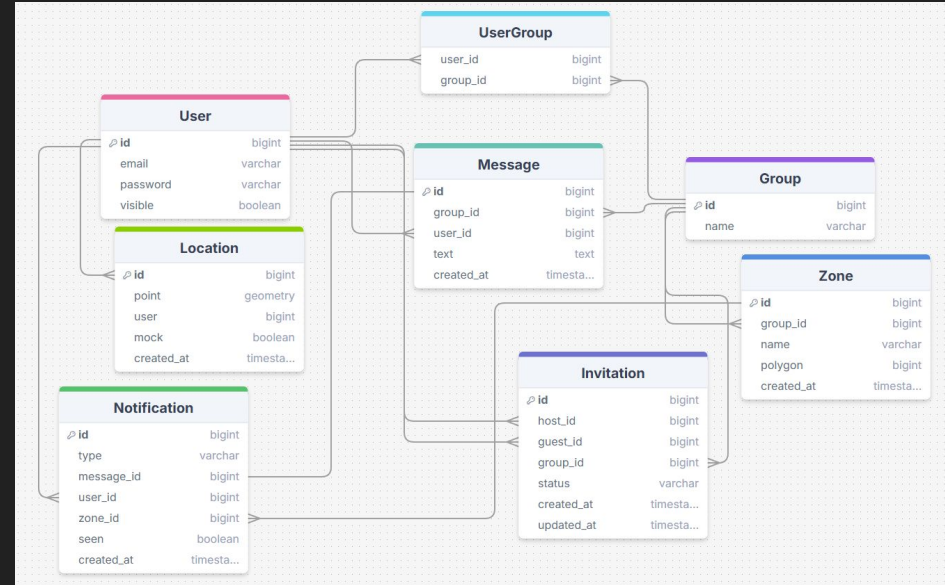
WireGuard Service / UI / Mobile app

- Secure end to end connection / tunnels
 - Images available (Service and UI)
 - Easy configuration from UI
 - Clients config share:
 - QR Code
 - Configuration file
 - Email
 - Telegram
 - Global: Updates changes from UI
 - Easy configuration on mobile app
 - QR Code read
 - Configuration file import
 - Traffic routing definition from UI
-
- Access WireGuard UI through Ngrok
 - Encrypt data with WireGuard mobile app
 - Decrypt data with WireGuard service
 - Route data to backend service



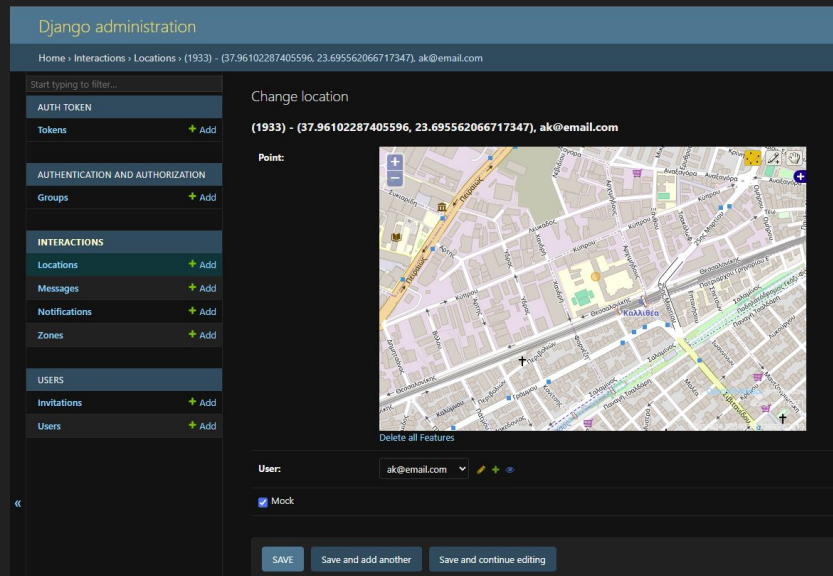
Backend (PostgreSQL)

- Free, open source
- Reliable (28 years since initial release 1996)
- Fast, Scalable
- Docker images
- Extensions support (images available)
 - Timescale (IoT applications, chunked data - Hypertables)
 - PostGIS (geospatial data, special functions)
- Determine zone entrance / exit with PostGIS



Backend (Django)

- Full stack (Jinja templates / template inheritance)
 - Excellent documentation
 - OOP practices (source code $\Rightarrow$ learning source)
 - Built in functionality
 - Intuitive ORM
 - Admin support
 - Easily integrates with PostgreSQL
 - Built in support for PostGIS extension with postgres backend
 - Geodjango (geographic framework)
 - Built in ORM support for geospatial queries
 - Built in geospatial model design support with geodjango (polygon-fields, point-fields)
 - Built in admin support for geospatial data
 - Variety of libraries (e.g. Django channels)
 - Steep learning curve (lot to learn)
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- Data modelling
 - Authentication
 - Determine zone entrance / exit with PostGIS
 - Notification handling (signals)
 - Webhook trigger (signals)



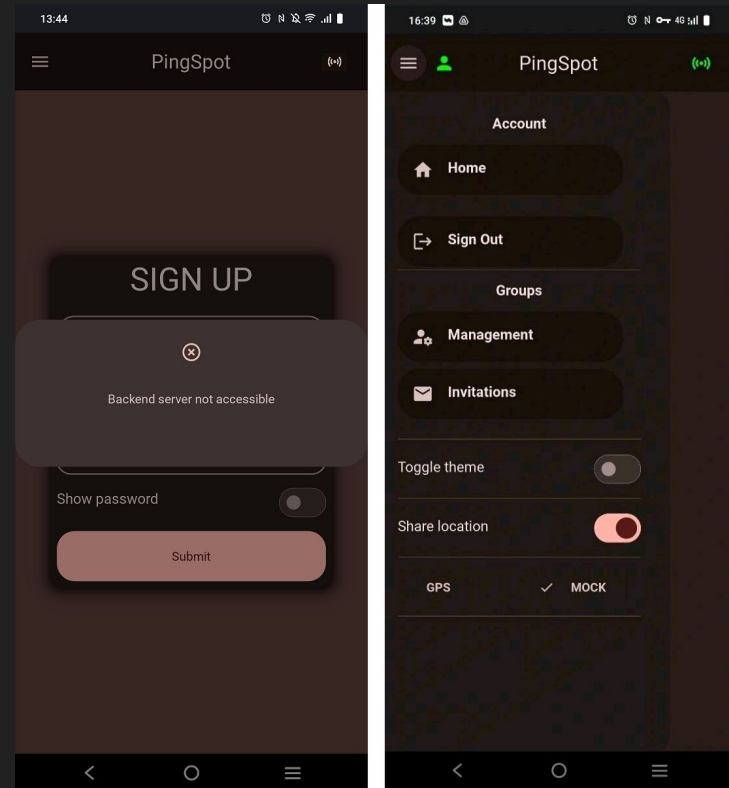
Backend (Django Rest Framework)

- Standard for building Web APIs
- Built in functionality
 - Token Authentication
 - (De)serialization
 - Validation
- Built in Browsing API/documentation tool
- Geospatial data support
(*djangoRESTframework-gis*)
- Great choice for microservices architecture

```
{
  "id": 1,
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [
      23.6983667,
      37.9623023
    ]
  },
  "properties": {
    "email": "ak@email.com",
    "mock": true,
    "created_at": "2024-12-29T19:29:51.463228Z",
    "user": 1
  }
}
```

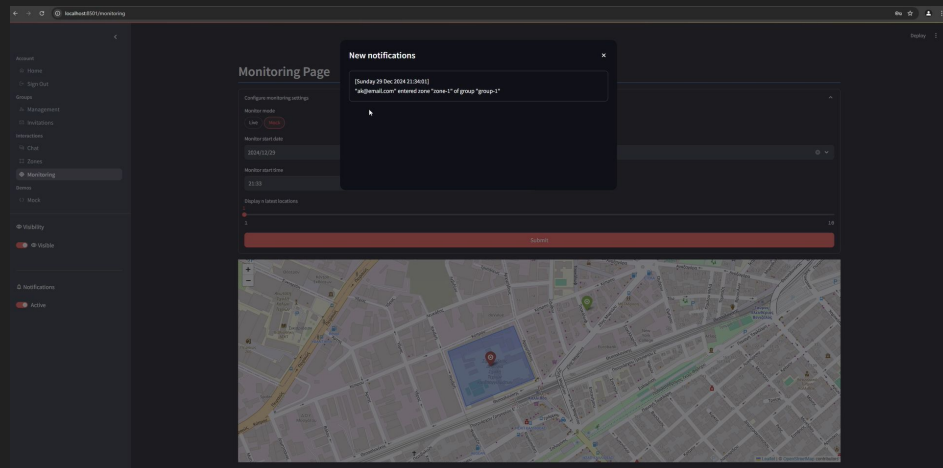
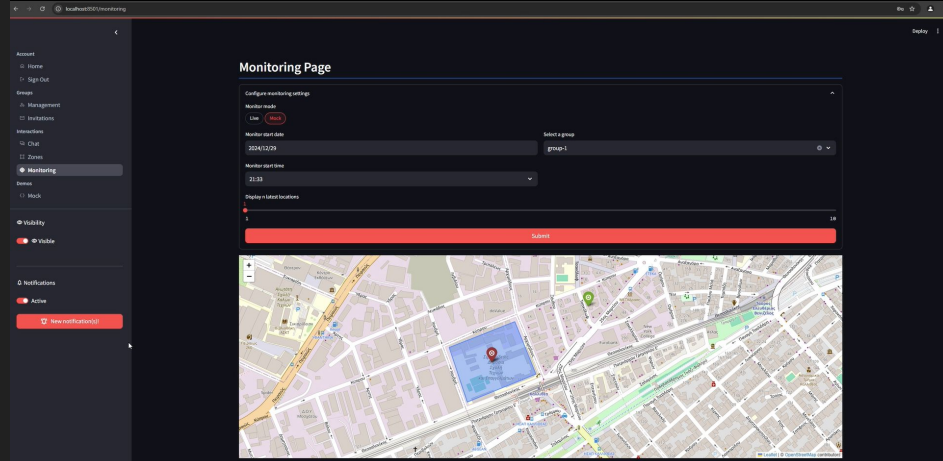
Mobile Application (Kivy / KivyMD)

- Kivy language
 - Declarative style UI style
 - Separation of concerns (interface vs logic)
 - Descent option for sticking with Python
 - Moderate learning curve
 - KivyMD
 - Material design widgets
 - Native look and feel
 - Buildozer (apk building)
 - Obscure *.spec* file handling
 - Hard dependency management
 - Limited and conflicting resources
 - Beautiful results / unlimited patience trade-off
-
- Backend service connection through WireGuard mobile app
 - Send GPS location data



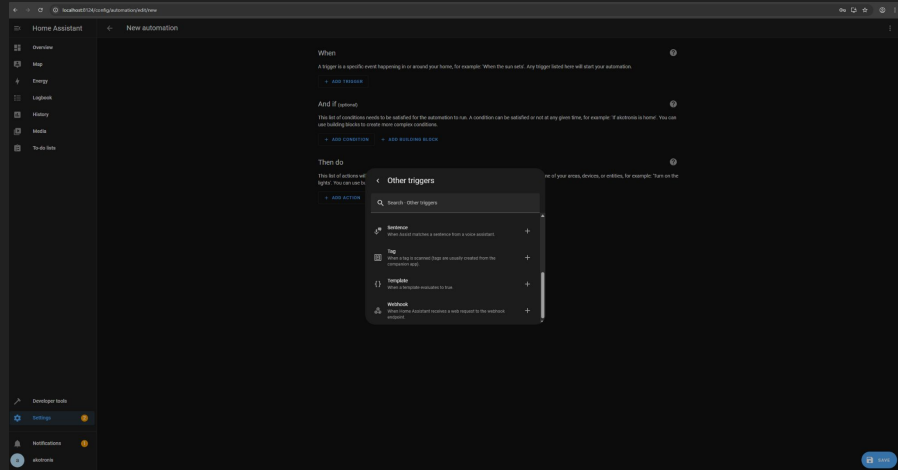
Frontend (Streamlit)

- Pros
 - No HTML/CSS
 - Variety of widgets
 - Beautiful layout (mostly responsive)
 - Custom "components"
 - Large community
 - Rapid prototyping
 - Recently added support for auth providers (keycloak etc)
- Cons
 - Page reruns on interactions (except for *fragments*)
 - Limited reusable component support
 - Workarounds needed for real time communication
- Alternatives
 - PyScript, Pyodide (WebAssembly)
 - Mercury, PyGWalker (Tableau like, Notebooks to webapps)
 - Panel, Taipy
 - Flet, NiceGUI
 - Reflex, FastHTML
- Authentication
- Group management
- Group invitations
- Zone definition
- Location monitoring (polling)
- Notifications (polling)



Home Assistant

- Widely used open-source home automation platform
- Thousands of devices supported (TP-Link, Zigbee)
- Automation rules, scripts, dashboards yaml configurations
- Runs on Raspberry Pi, VM, Docker, no cloud dependencies
- Device discovery (auto)
- Supports some implemented features (device monitoring, geofencing, WG addon)
- Webhook trigger definitions
 - `api/webhook/<tapo-webhook-turn-on-id>`
 - `api/webhook/<tapo-webhook-turn-off-id>`



Live Demos / Github Repo

- [Group management](#)
- [Group invitations](#)
- [Zone definitions](#)
- [Mobile application](#)
- [Flow without Tapo trigger](#)
- [Flow with Tapo trigger](#)
- [Thesis Github repo](#)

Questions and Answers

